Mathematical AI Summary

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Date: October 18, 2026

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This is an AI's attempt to rewrite GR in the way I envision it. Since I have absolutely no clue about the equations (I am a layman, not a physicist), I don't know how well it worked out. But perhaps it clarifies how I imagine it... The goal was that wherever GR speaks of mass/matter, metric, curvature, it is calculated using the energy equivalent and named/referenced accordingly with respect to energy.

The AI response:

I would structure this section across three levels: first, the energy form of classical gravitation, then GR, and finally your interpretation.

1. Expressing Mass as Energy

From the equivalence of mass and rest energy, it follows that:

- **E_0 = m * c^2 <=> m = E_0 / c^2**

Where **E_0** is the rest energy and **c** is the speed of light.

The Newtonian gravitational force

- **F = G * (M * m) / r^2**

can therefore be rewritten as:

- **F = G * (E_0,1 * E_0,2) / (c^4 * r^2)**

For the acceleration of a test body, this yields:

- **a = G * E_0,1 / (c^2 * r^2)**

Here, **E_0,1** represents the rest energy of the causing body, such as Earth.

Significance for your hypothesis: In this approximation, the gravitational effect can be expressed entirely using energy quantities instead of mass quantities. The mathematical prediction remains identical.

2. Formulating GR with Respect to Energy

The Einstein field equation reads:

- **G_mu_nu = (8 * pi * G / c^4) * T_mu_nu**

Where:

- **G_mu_nu:** describes spacetime geometry and its curvature.
- **T_mu_nu:** describes energy and momentum distribution, including energy density, momentum flux, pressure, and stresses.

- **G:** gravitational constant.
- **c:** speed of light.

Crucial for your hypothesis: GR already contains energy in its fundamental equation. Matter is not treated as the sole source of gravitation in this description.

In the limit of weak gravity and slow motions, the relationship with energy density **u** can be illustrated as:

- **Laplace(Phi) ≈ 4 * pi * G * (u / c^2)**

Here, **Phi** is the Newtonian gravitational potential and **u** is the energy density. This approximation assumes that rest energy dominates and that pressure and other relativistic contributions are negligible.

The equation shows how energy density contributes to the gravitational effect in this limiting case. However, it does not state that full spacetime curvature is determined solely by local energy density in all situations.

3. Interpretation of Your Hypothesis

Your approach can thus be summarized as follows:

1. **Ontological assumption:** Matter is not a substance fundamentally separate from energy, but can be understood as a bound state of energy.
2. **Mathematical approach:** Mass quantities can be replaced by their energy equivalents in suitable equations. In GR, energy and momentum distribution is already described by **T_mu_nu**.
3. **Interpretation of gravity:** Gravitational effect is consistently viewed from the perspective of energy distribution and dynamics. Whether spacetime geometry is fundamental in this context or possesses a deeper physical description remains an additional question of interpretation.

An electromagnetic field does not need to be introduced as an additional quantity for this purpose initially. In GR, an electromagnetic field itself contributes to the energy-momentum tensor. However, it does not follow that spacetime curvature is identical to an electromagnetic field.